

CSE7101- Capstone Project
Review-1

Udyog Saarthi App - (Progressive Web-based Application) for Adults undergoing Job coaching for opportunities under 4% reservation in NIEPMD and other Institutions.

Batch Number: CCS_09

Roll Number

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Problem Statement Number: PSCS_39

Organization: Ministry of Social Justice and Empowerment

Category (Hardware / Software / Both): Software

Problem Description:

Despite progress, about 64% of persons with disabilities (PwDs) in India are unemployed. A key reason is that a vulnerable group, eligible for a 4% reservation in government jobs, often misses these opportunities because they cannot access or understand complex job notifications from traditional media. Their caregivers frequently lack the awareness to help them apply. As a result, many reserved positions go unfilled, undermining the goals of the Rights of Persons with Disabilities Act, 2016, and preventing PwDs from achieving economic independence and social inclusion.



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Problem Statement

- **High Unemployment:** In India, approximately **64% of persons with disabilities** (PwDs) are unemployed, particularly those with intellectual and developmental challenges.
- **Information Gap:** A major cause is the lack of accessible, timely, and simplified information regarding job opportunities under the **4% reservation quota** mandated by the Rights of Persons with Disabilities Act, 2016.
- **Accessibility Barriers:** Government and PSU job notifications are often complex, filled with jargon, and scattered across numerous websites, making them inaccessible to the target user group and their caregivers. This results in a significant number of reserved positions going unfilled.

Objectives

- **To Centralize Information:** To design and develop a single, unified mobile platform that aggregates all relevant government and PSU job vacancies reserved for PwDs.
- **To Enhance Accessibility:** To simplify and translate complex job notifications into easy-to-understand language using AI, and to build the app with accessibility features like Text-to-Speech, high-contrast mode, and multi-language support.
- **To Empower Users & Caregivers:** To create dedicated modules for both PwD users and their guardians, providing tools for profile management, application guidance, and access to vocational resources.
- **To Improve Vocational Readiness:** To integrate skill-building tools like a mock interview simulator and an AI chatbot to help users prepare for the employment process and build confidence.

Background and Related Work

- **Legislative Context:** The **Rights of Persons with Disabilities Act, 2016** is a landmark law that mandates a 4% reservation in government jobs for specific categories of disabilities. However, its implementation is hindered by a persistent information gap.
- **Existing Platforms:** While general job portals like the **National Career Service (NCS)** exist, they are not specifically tailored to the unique cognitive and accessibility needs of individuals with intellectual or developmental disabilities. Other specialized portals are often limited in scope or lack the automation to stay consistently updated.
- **Identified Gap:** There is a clear need for a dedicated, user-centric application that not only lists jobs but also provides end-to-end support from understanding the opportunity to preparing for the interview specifically for this underserved community.



Analysis of Problem Statement

Technology Stack Components:

- **Frontend (Web App):** React.js (to build a progressive web app - PWA)
- **Backend (API Server):** Node.js with Express.js
- **Database:** MongoDB (NoSQL)
- **Web Scraping Service:** Python (using BeautifulSoup, Requests)
- **AI & ML Integration:** Google Gemini API (for chatbot and text simplification)
- **Deployment & Hosting:**
 - **Frontend:** Netlify
 - **Backend:** Render/Vercel
- **Containerization & Deployment:** Docker

Analysis of Problem Statement (contd...)

Software and Hardware Requirements:

- **Software:** VS Code, Git/GitHub, Postman, Node.js, Python 3.10, Modern Web Browser (Chrome/Firefox), Docker Desktop.
- **Hardware:** Intel i5 / AMD Ryzen 5 Processor, 8 GB RAM (16 GB Recommended), 256 GB SSD.

Cost Analysis (Projected):

- **Recurring Costs (Monthly): ₹0.** We will utilize the free tiers of cloud providers like Render/Vercel (for web hosting) and MongoDB Atlas (for the database), which are sufficient for development and initial deployment.
- **Potential Future Costs: \$25 (~ 2100 INR) one-time fee** for a Google Play Developer Account would only be required if the project expands to a native mobile app in a later phase.

Analysis of Problem Statement (contd...)

User Challenges:

- **Cognitive Overload:** Complex language and cluttered layouts on official websites are significant barriers.
- **Information Discovery:** Job listings are fragmented across dozens of websites, making them difficult to track.
- **Lack of Guidance:** Users and caregivers often lack the knowledge to navigate the complex government application process online.

Proposed Solutions:

- **AI-Powered Simplification:** An AI model will automatically summarize and re-phrase complex job descriptions into simple, accessible language.
- **Automated Data Aggregation:** A dedicated web scraping service will continuously gather and update job listings into one centralized platform.
- **Guided Workflows & Resource Hub:** The web app will provide step-by-step application guides and a directory of coaching centers (like NIEPMD).

Innovation or Novel Contributions

- **Dual-User Model (PwD & Caregiver):** The platform is uniquely designed with a dedicated, interconnected portal for caregivers, empowering them with tools and information to support the job seeker.
- **AI-Powered Content Simplification:** Unlike traditional job portals, Udyog Saarthi will use AI to automatically "translate" complex government jargon into simple, accessible language on a web platform.
- **Integrated Skill-Building Ecosystem:** The web app moves beyond being an information portal by integrating a **mock interview simulator** and an **AI chatbot**, creating a holistic support system.
- **PWA for Accessibility:** Building as a Progressive Web App ensures it is instantly accessible on any device with a browser (desktop or mobile) without needing an app store, and can be "installed" on a home screen for easy access.

Github Link

<https://github.com/holycarrots/Capstone-Project-2025>

Github Link



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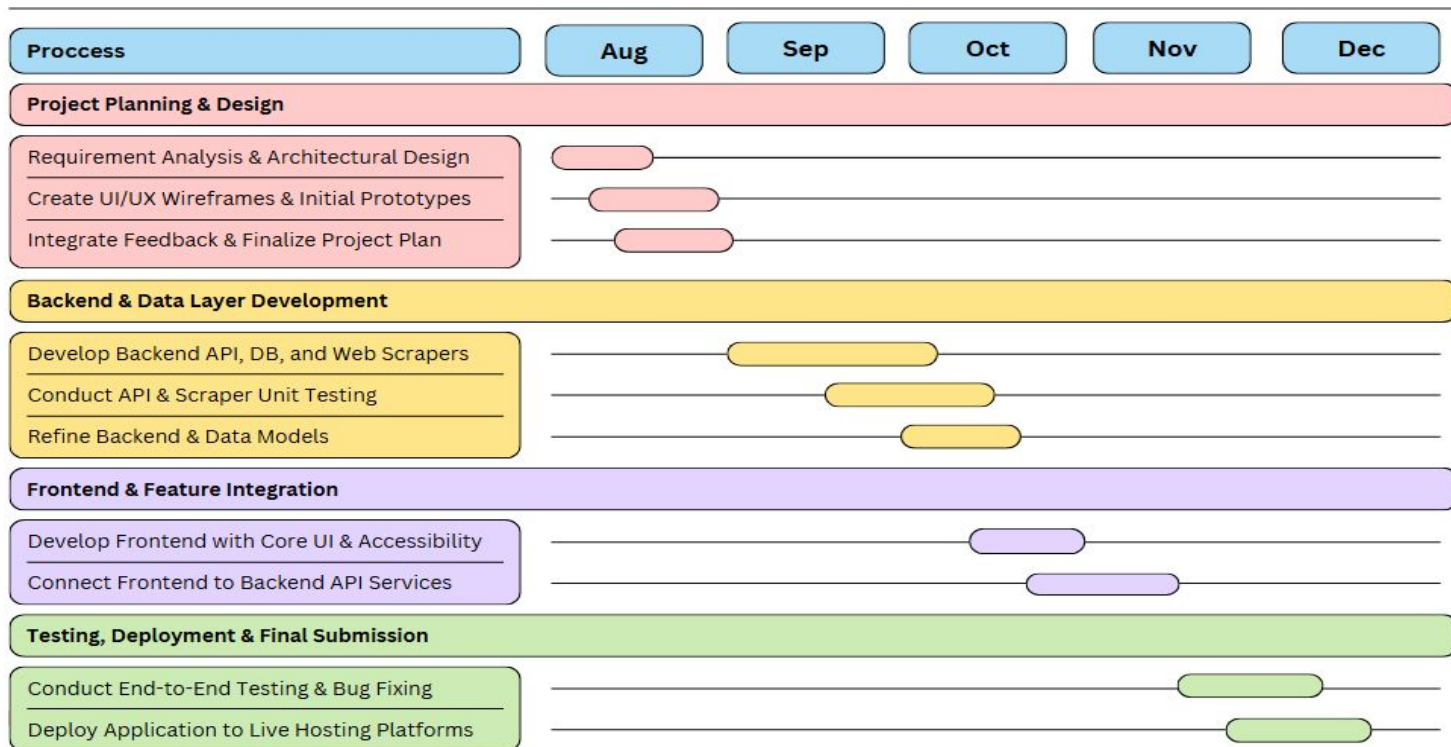
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Timeline of the Project (Gantt Chart)

Udyog Saarthi App

Gantt Chart



References (IEEE Paper format)

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- [7] S. V. Anjikhane and S. A. Doddi, "Udyog Saarthi: Enhancing Employment for Persons with Disabilities."



Thank
You!

